Cyber Security Strategies for Financial Organizations: A Consultant's Report for Global Finance Inc.

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Executive Summary

This 92's digital solutions consultancy research assesses the cybersecurity stance of Global Finance Inc., analyzing the current cyber threat environment, strategic implementations of Cyber Threat Intelligence (CTI), and conformity with international cybersecurity frameworks. Also contains a holistic assessment of vulnerabilities, access controls, and adherence to legal and ethical principles to formulate a pathway for financial sector cyber resilience.

*Please refer to Appendix A for short abbreviations familiarity

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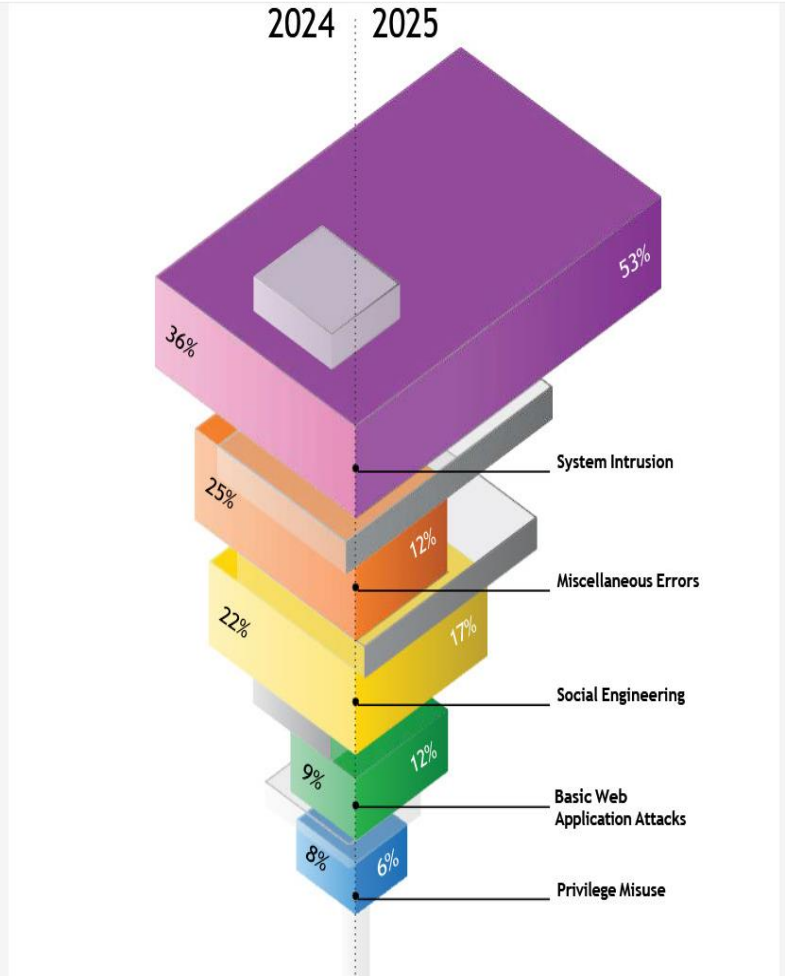
1. The Landscape of Cyber Threats and Cyber-Crime

1.1 Introduction

Key infrastructure, including government defense, technology providers, healthcare, and educational services, has become a major target for cybercriminals, particularly financial groups, which ranked second in the 8isoft ranking (2024). Figure 1 illustrates the increase in percentile comparison of 2024 and 2025 threat attacks, with system intrusion being the most significant, with 53% attack types for the Financial sector due to their greater reliance on technology than before, creating greater risks for cyberattacks. With deviation in motive assentive, these are commonly sought out by these threat actors, ranging from:

- Nation-State
- Cyber Criminals
- Hacktivists
- Insider Threats
- Script Kiddies
- Terrorist Groups
- Competitor Espionage
- Thrill Seeker



See Appendix for definitions

Figure 1: Verizon "Data Breach Investiagtion Report 2025

According to Up Guard (2024), the financial sector's gigantic management of valuable economic assets makes them increasingly vulnerable to attacks for monetary gain. Global Finance Inc. must acknowledge cyber threats and landscapes to enhance defenses against sophisticated attack methods across all attack surfaces.

1.2 Industry trends and Landscape threats

According to CrowdStrike (2024), the common threat types associated with financial sectors are:

Threat Type	Mobile Phones	Desktops	IoT Devices	Targets Banking Apps/Webs /Networks/Clouds	Other Notes / Statistics
Spyware & Ransomware	✔	✔	✔	✔	• 97% of mobile threats affect Android. Only 3% affect iOS. Silent Breach (2025) • Ransomware used for Cryptojacking on CPUs • SQL injection (OSWAP) is the dominant attack on bank webpages. Waratek (2025)
Banking Trojans & Botnets	✔	✔	✔	✔	AP news (2023) Qakbot example cost $58 million in theft from Trojan bots.
Phishing	✕	✔	✕	✔	Commonly targets users via fake banking sites/emails.

As Agatha Christie, the detective author, stated, “A crime without a motive is a useless puzzle.” Likewise, diverse threat actors conduct cybercrimes driven by their own motives, which in turn influence their attacks. As explained below:

1.3 Money as the Primary Motive: Targeting Financial Institutions

Figure 2:

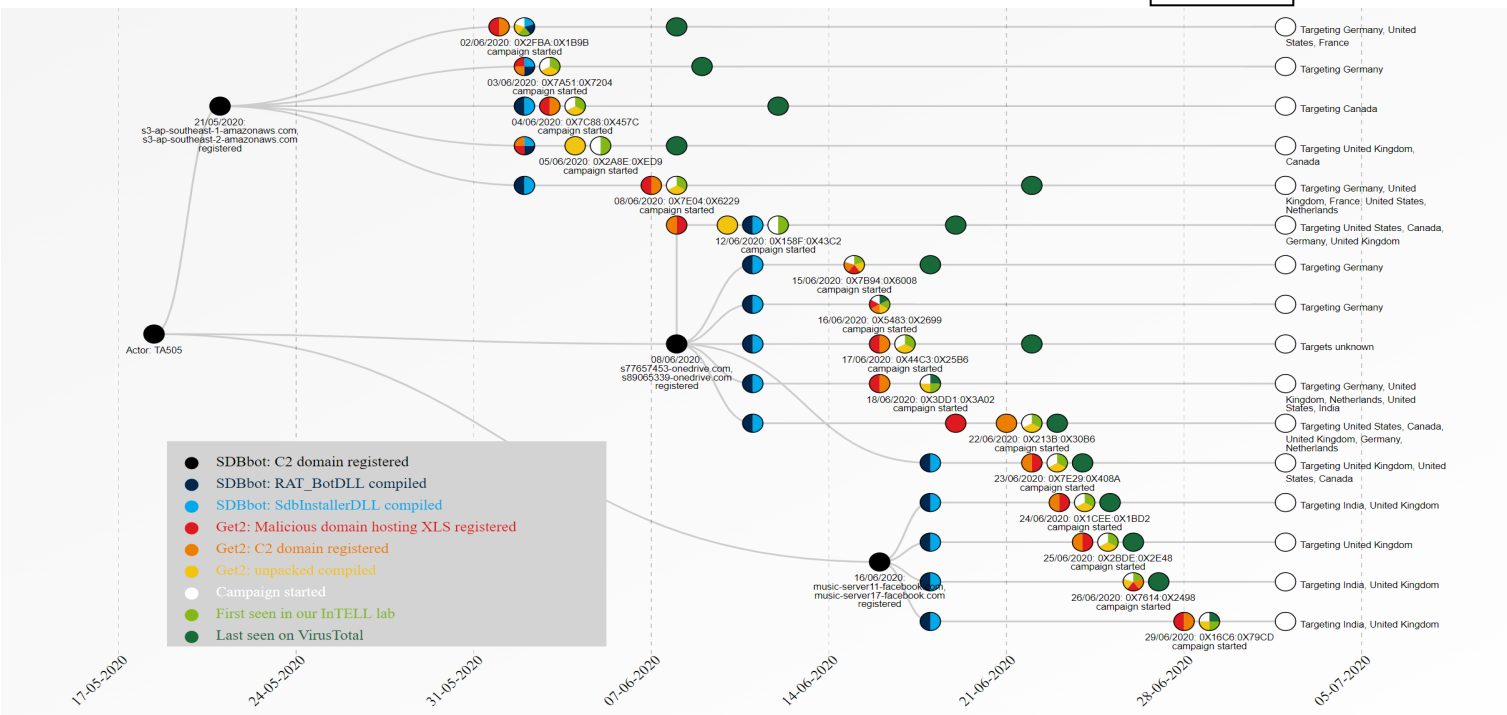


Figure 2, sourced from Antonis Terefos, illustrates **1.2** malware, including the Remote Manipulator System backdoor used by TA505 to target global financial institutions. The AMADAY malware family from various nations from May to February 2020 (NCC Group, 2025), their signature move being phishing (B.E.C. included), exploited poor cybersecurity awareness to capture login credentials and distribute 'Dridex' ransomware, encrypting data for ransom.

Impacts**:**

- Financial Loss - U.S. Treasury (2019): TA505 campaigns caused over $100 million in losses, likely earned millions in the December 2020 operation
- Loss of client trust
- regulatory penalties
- Increased Security cost

1.4 Motivation Besides Capitalism

1.4.1 Hacktivism in the Financial Sector

According to BlackFog (2025), Cyberattacks may arise from protest rather than profit, as hacktivists target banks over savings rates, unethical investments, and environmental issues, utilizing DDoS attacks to disrupt operations and highlight injustices.

1.4.2 Political Warfare in the Financial Sector

The 2007 Estonia cyberattacks, as reported by Thematic Arena, linked to Russian involvement, targeted banks and disrupted digital services—sometimes preventing customers from viewing their balances. The attacks undermined public trust, triggered riots, recession, currency instability, and market distortion.

1.4.3 “Show Off” Hack: TalkTalk Data Breach

Published by The Standard newsletter, by Francesca Gillett (2016), The 2015 TalkTalk breach by a 15-year-old used SQL injection, exposing over 150,000 bank details and costing TalkTalk £60 million in damages and £400,000 fine (Data Leaks Lawyers,2025), illustrating significant damage from low-skilled individuals by motivation sourced from social recognition from fellow peers.

1.4.4 Insider Threat

Insider threats in the financial sector occur when employees or contractors misuse privileged access to leak sensitive data, maliciously / negligently, often due to motives of retaliation/ careless negligence. A Ponemon Institute study found that 65% of departing employees took proprietary data, risking their former company's competitiveness and contributing to potential data breaches, as reported by Cyfor Forensics.

1.5 Protection Advisory Plans

As AI exacerbates phishing and vulnerabilities, it also intensifies the conflict between threat actors and cybersecurity. The introduction of Initial Access Brokers also creates complex threats like Ransomware-as-a-Service, with data breaches averaging $4.88 million in 2024 (IBM report), underscoring the need for stronger cybersecurity. Global Finance Inc must align security measures with national standards and continuously monitor risks through integrated defense strategies. Furthermore, to counter diverse threats, the company must align defenses with attackers' incentives and tactics, implementing attack surface management strategies like network segmentation to limit lateral movement and multi-factor authentication (MFA) for secure access to banking webpages, thus decreasing the threat landscape.

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2. Strategic Importance of Cyber Threat Intelligence (CTI)

CTI helps with a proactive cybersecurity strategy. For financial institutions like Global Finance Inc., it provides not only situational awareness but also predictive defense capabilities. Broken down into four segments, it includes:

Figure 3:

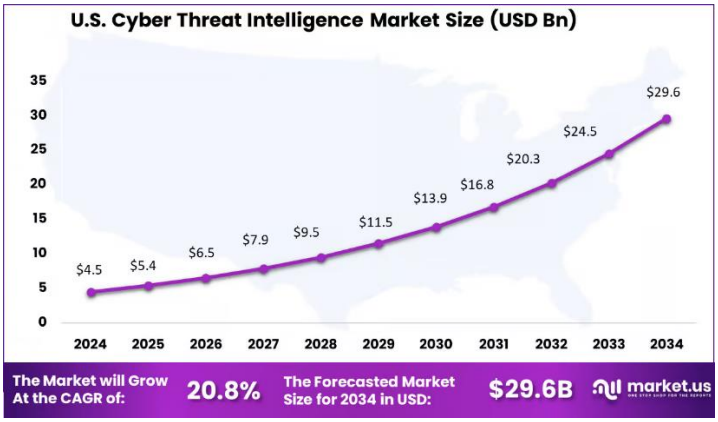

Figure 3: Statistics of cyber threat intelligence

- **Strategic CTI** -tailors' strategies regarding risks, investments, and threat landscape changes to CEOs
- **Tactical CTI** with SOC personnel it allows an understanding of adversary TTPs through frameworks like MITRE ATT& CK.
- **Operational CTI** enables real-time threat response that can be used by SIEM and SOAR systems
- **Technical CTI** provides threat data (IPs, hashes, domains).

Refer to appendix A; D; F for definitions of acronyms

2.1 Miscellaneous Tools

CTI is vital in mitigating phishing and **(1.1**) popular system intrusion threats through email detection, training, and breach monitoring. It employs TTPs, IOCs, and frameworks from platforms such as FS-ISAC and MISP to enhance defenses. Many CTI automation systems now use AI for proactive threat monitoring and early vulnerability detection, helping organizations like Global Finance Inc. stay ahead of evolving threats while improving response speed and reducing blue and red analyst workload.

2.2 AI in Security

Illustrated in Figure 3, the global Cyber Threat Intelligence (CTI) market is rapidly growing, projected to reach 91.7 billion by 2034, with a CAGR of 22.1% from 2025 to 2034 (Market.us, 2025). In light of this trend, Global Inc. should improve its cybersecurity by implementing a robust CTI Intelligence strategy, employing threat modelling, and regularly assembling purple teams—whether or not they are powered by AI—to enhance incident response, minimize attack surfaces.

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3. Review of Key Cyber Security Standards and Frameworks

According to Jay Ryerse (Connectwise,2024), the general recommended Cybersecurity frameworks providing Global Inc with an organized method for managing risk, ensuring compliance, and maintaining operational resilience, which in turn foster trust with stakeholders, are as follows:

3.1. ISO/IEC 27001 – Information Security Management System (ISMS)

According to ISO (2025), ISO/IEC 27001 is a global standard that helps organizations establish an effective ISMS, focusing on a risk-based approach to protect data, especially in finance. It involves risk assessments, security controls, and continuous

monitoring. Following these standards enhances data confidentiality, integrity, and availability, lowers breach risks, and ensures compliance with data protection laws, whilst certification validation fosters trust with clients.

3.2 NIST Cybersecurity Framework (CSF)

The NIST is a flexible model for managing cybersecurity risks, organized into five core functions:

- Identify
- Protect
- Detect
- Respond
- Recover

See appendix: E

This structure helps organizations understand assets and risks, detect intrusions, respond effectively, and restore systems. Its adaptability allows Global Finance Inc. to customize cybersecurity programs to its needs, aligning security with business objectives and enhancing communication between teams.

3.3 Summary

The appropriate framework combination depends on the international regulatory requirements of whom Global decides to do business and their goals. ISO 27001 is necessary for vendor trust and compliance through certification validation, while NIST suits U.S. regulations and its ability to assess its incident response readiness. Global Finance Inc. should adopt a hybrid approach, integrating ISO/IEC 27001 with NIST CSF agility and any complementary additional framework to ensure compliance, enhance cybersecurity, and build stakeholder trust while addressing vulnerabilities.

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4. Assessment of Common Vulnerabilities and Access Control Mechanisms

4.1 Vulnerability Management

Vulnerability management is an essential process for securing financial systems. Focusing on identifying, assessing, remediating, and monitoring security flaws in networks and applications, thus enhancing attack surface management (mentioned in **1.5**) by minimizing exploitable weaknesses. A widely recognized organization on application security is OWASP, known for its Top 10 list, which highlights critical risks such as injection flaws (e.g., SQL injection, mentioned in **1.2**) and broken access control. However, many publicly disclosed vulnerabilities are catalogued in the (CVE) system, where high-impact exploits such as EternalBlue—one of the most significant—are documented

4.1.1 Eternal Vulnerabilities Examples

Figure 4:

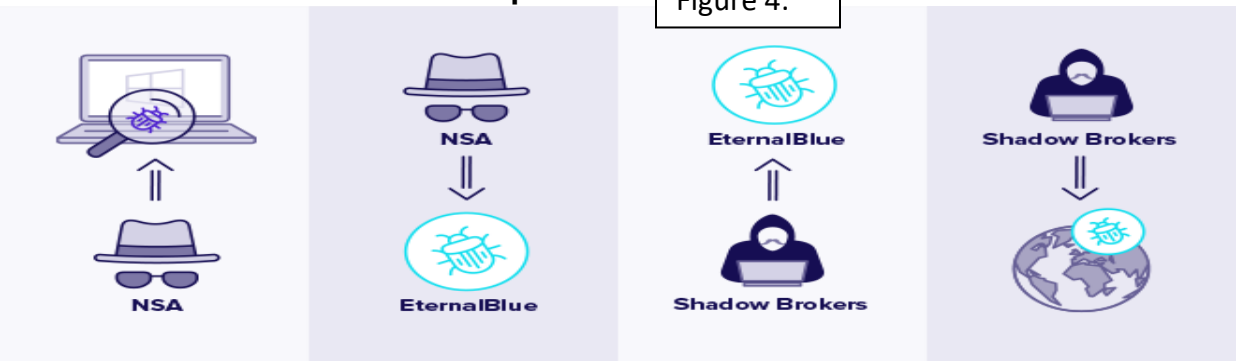


Figure 4 shows chronological events from the development of the NSA's EternalBlue to exposure to the “Shadow Brokers.”

(Tenable 2025) EternalBlue MS17-010 CVE-2017-0144 (VPR score = 9.8) affects Windows OS, targeting the SMBv1 file-sharing protocol, making it vulnerable to ransomware and cyberattacks.

Figure 5:

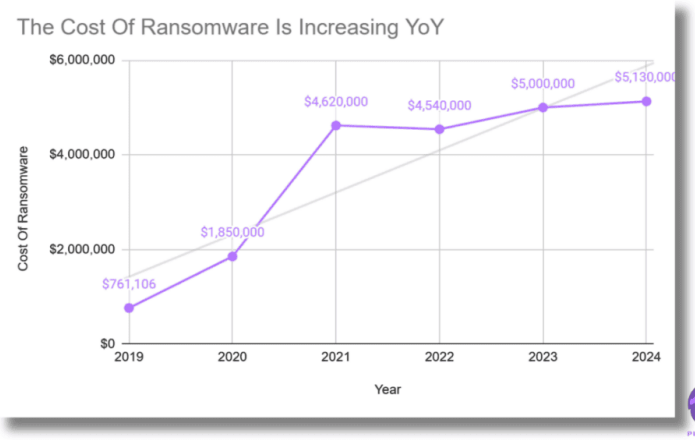
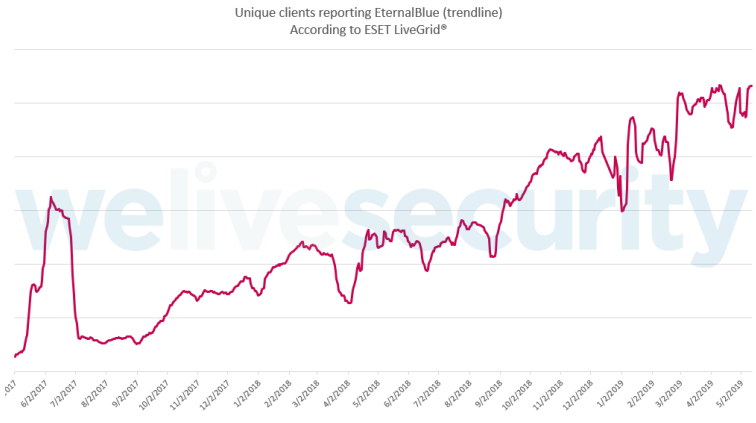


Figure 6:



Both diagrams show a positive correlation between ransomware costs and recorded EternalBlue reports, with estimated global losses of US$4 billion (Kaspersky, 2025) from the WannaCry attack leveraging the EternalBlue exploit.

4.2 Access Controls

Access control regulates those who can use resources in an organization, ensuring only authorized users can access specific data or functions based on permissions.

4.3 Access Control Mechanisms Recommendations:

- Role-Based Access Control (RBAC): Access is granted based on a person's role within the organization. It supports auditing and helps ensure people only have access to what they need.
- Attribute-Based Access Control (ABAC)- decisions are made based on different attributes—like who the user is, what they're trying to access. It's more flexible and allows for better access control.

According to NIST's "Guide to ABAC (2014)" report, other models like DAC and MAC (Appendix D) are often avoided due to being overly flexible or too rigid for commercial use, making them better suited for military or government systems.

4.4 Methods of Mitigation

Routine patching, configuration management, and penetration testing should address vulnerabilities, including OWASP SQL injection, that will allow 'least privilege' modifications of access controls if needed. Benefits include reduced risk, enhanced compliance with regulations, and improved incident response strategies as financial firms handle sensitive customer data and face stricter regulations.

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5 Legal, Social, Ethical, and Professional Issues

(Cybersecurity involves complex legal, ethical, and professional responsibilities.)

5.1 Legal

- **Regulatory Framework**: Global Finance Inc. needs to stick closely to laws like the GDPR, UK DPA 2018, and CMA 1990 for user data protection and avoid fines.
- **Jurisdictional Risks**: Lack of a U.S. federal privacy law creates challenges for international businesses; Global Finance Inc. should exercise caution in engaging foreign businesses.
- **Data Governance**: Global Finance Inc must address data privacy, intellectual property rights, and transparency in all partnerships.
- **Ethical Balance**: The Company's cyber professionals need to balance organizational protection with individual privacy and legal compliance.
- **Public Trust**: As cyberattacks damage trust and economic stability, prompting the company is prompted to invest in digital literacy and client transparency.

5.2 Professional & Ethical

- **Ethical Dilemmas**: Cybersecurity experts could face complex decisions, especially when discovering unethical or illegal activities.
- **EternalBlue**: Section **4.1.1** highlights the thin line between surveillance and privacy and the consequences of breached data.

- **System Design**: SOUPS 2024 Insight conference - warns of the negative impact of poor security practices on user behaviour. Raising the question: should users be punished for errors, or should systems be more user-friendly?
- **Human Error & Threats**: Linked to Insider threats (**1.1.4)** and (**1.2)** phishing; emphasizes the need for ethical, user-centred design, as people make mistakes, so security systems should support users, not punish them.
- **Consequences**: Cyberattacks carry legal, reputational, and financial costs; an undesirable phenomenon for Global Finance Inc., thus the importance of resilience strategies to mitigate.
- **Resilience Measures**: The company should prioritize multi-factor authentication, zero-trust networks, and staff training for protection.

5.3 Professional Conduct in Hacking & AI

- **Case Example**: Matt Johansen's YouTube video " Is an AI really the top hacker in the US right now?" explores AI ethical hacking and the misuse of exploits like EternalBlue.
- **Disclosure**: Concluding that findings must be reported promptly to reduce risk and prevent exploitation.
- **AI in Security** (linked to **2.2**): When using AI in cybersecurity, Global Finance Inc. should stick to ethical guidelines from organizations like e.g. EC-Council. That means keeping confidentiality, integrity, and accountability—for both humans and machines.

5.4 Social

- **Incident Response:** Global Inc. needs to communicate clearly before, during, and after a cyber incident—especially to manage dilemmas like the one in section **1.4.2**.
- **Ethical Standards:** How the company communicates should follow professional ethics from groups like BCS, ISACA, etc.
- **Internal Communication:** Staff should be trained to tackle social engineering tactics for user data protection and the company's reputation.
- **External Communication:** Being honest with clients builds trust. Global Inc. should enhance customer digital awareness to prevent confusion, reduce legal risk, and mitigate incidents.

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6. Conclusion

Global Finance Inc. must enhance its cyber defence by incorporating CTI, enhancing staff awareness, and ensuring compliance with ISO and NIST standards.

Understanding TTPs of common threat actors, Management of weaknesses and attack surfaces, and responsible data handling are crucial. Fostering a culture of cybersecurity resilience through ongoing education and strategic planning is vital for protecting financial systems from emerging threats.

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7. Appendix

Table of Figures:

Figure 1: Verizon “Data Breach Investigation Report 2025

Figure 2: Dataset 2, Campaigns overview sourced by Antonis Terefos

Figure 3: Statistics of cyber threat intelligence

Figure 4: Cyber Threat Intelligence Threat Global Market Report 2025

Figure 5: The Cost of Ransomware is Increasing YoY

Figure 6: EternalBlue Reports Trendline

Appendix A: Key Abbreviations and Concepts

- SOC – Security Operations Center
- GDPR – General Data Protection Regulation
- UK’s DPA 2018 – United Kingdom’s Data Protection Act 2018
- CMA 1990 – Computer Misuse Act 1990
- CIA Triad – Confidentiality, Integrity, and Availability
- NIST – National Institute of Standards and Technology
- BEC (Business Email Compromise) – Occurs when cybercriminals impersonate trusted figures (e.g., CEOs) to trick employees into sending funds or sensitive data. Microsoft Reference: https://www.microsoft.com/en-gb/security/business/security-101/what-is-business-email-compromise-bec
- ISACA and (ISC)² Code of Ethics –

  - ISC²: Offers the CISSP certification; focused on training cybersecurity professionals.

  - ISACA: Offers the CISM certification; focused on governance, auditing, and risk management.

Appendix B: Threat Actors

- Nation-States – Government-backed cyber espionage, warfare, or sabotage.
- Cyber Criminals – Profit-driven actors involved in data theft, ransomware, etc.
- Hacktivists – Ideologically motivated attackers promoting social/political causes.
- Insider Threats – Current or former employees/contractors abusing internal access.
- Script Kiddies – Inexperienced individuals using pre-built hacking tools.
- Terrorist Groups – Use cyberattacks to spread fear or disrupt infrastructure.
- Competitor Espionage – Corporate spying to gain unfair market advantage.
- Thrill Seekers – Individuals hacking for excitement or curiosity.

Appendix C: Common Malware Types

- Ransomware – Locks or encrypts files, demanding payment for release.
- Phishing – Fraudulent emails/messages aimed at stealing credentials or data.
- Banking Trojans – Target financial data through deception.
- Spyware – Covertly collects user data and behavior.
- Botnets – Networks of compromised devices controlled remotely for attacks.
- Code Injections – Malicious code injected into software/systems.
- DDoS (Distributed Denial of Service) – Overwhelms systems with excessive traffic.

Appendix D: Threat Intelligence & Access Control

Threat Intelligence Tools and Standards

Reference: https://www.cloudflare.com/en-gb/learning/security/what-is-stix-and-taxii/

- FS-ISAC – Financial Services Information Sharing and Analysis Center.
- MISP – Malware Information Sharing Platform.
- STIX/TAXII – Standards for structured threat information sharing.
- MITRE ATT&CK – Knowledge base of adversary tactics and techniques.
- SIEM vs SOAR – SIEM: Threat detection/log analysis; SOAR: Automated response.

Access Control Models

- RBAC (Role-Based Access Control) – Access based on job roles; promotes least privilege.
- ABAC (Attribute-Based Access Control) – Access based on contextual attributes.
- DAC (Discretionary Access Control) – Resource owners assign access permissions.
- MAC (Mandatory Access Control) – The Central authority defines access based on classification.

Appendix E: NIST Cybersecurity Framework

- Identify – Recognize assets, risks, and business context.
- Protect – Implement security measures and safeguards.
- Detect – Monitor and identify cybersecurity incidents.
- Respond – Take action to contain and resolve incidents.
- Recover – Restore capabilities and services post-incident.

Appendix F: Threat Modeling Frameworks

- STRIDE – Security model highlighting six common threats:

  - Spoofing
  - Tampering
  - Repudiation
  - Information Disclosure
  - Denial of Service
  - Elevation of Privilege

- PASTA – Process for Attack Simulation and Threat Analysis.

  Focuses on risk-centric, attacker perspective-driven threat modeling.

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